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Inventor(s)	Hendriks; Rob Jacob et al.

Transferring viscous materials

Abstract

A method and device are described to transfer a viscous functional material onto a receiving substrate. A plate is provided having a cavity surface that includes a cavity. A plurality of individually addressable resistive heater elements are provided that are in thermal contact with respective zones of the cavity. Viscous functional material is provided in the cavity with a material composition that, when sufficiently heated, generates a gas at an interface between the cavity surface in the cavity and the functional material, to transfer the functional material from the cavity by the gas generation onto the receiving substrate. Respective portions of the viscous functional material in respective zones of the cavity are heated by supplying respective ones of the plurality of individually addressable heater elements with an electric power having a respective time dependent magnitude.

Inventors:	Hendriks; Rob Jacob (Eindhoven, NL), Bruning; Fabien Bernard Jacques (Eindhoven, NL)
Applicant:	Nederlandse Organisatie voor toegepast-natuurwetenschappelijk onderzoek TNO ('s-Gravenhage, NL)
Family ID:	70682635
Assignee:	Nederlandse Organisatie voor toegepast-natuurwetenschappelijk onderzoek TNO ('s-Gravenhage, NL)
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Primary Examiner: Campbell; Thor S

Attorney, Agent or Firm: Leydig, Voit & Mayer, Ltd.

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This patent application is a U.S. National Phase of PCT International Application No. PCT/NL2021/050309, filed May 12, 2021, which claims priority to European Application No. 20174175.8, filed May 12, 2020, which are both expressly incorporated by reference in their entirety, including any references contained therein.

FIELD OF INVENTION

(2) The invention relates to a method of transferring a high viscosity functional material, such as an electrically conductive or an electrically insulating paste, on a substrate.

(3) The invention further relates to a device for transferring such a material.

(4) The invention still further relates to a record carrier comprising instructions that when executed by a programmable processor causes an apparatus to perform the method.

BACKGROUND

(5) Interconnection of discrete bare chip components or microelectronic components (chips) on a substrate is a process for which precision material deposition with sufficient accuracy is necessary to allow for miniature electrical or other type of interconnects so that when a chip component becomes attached to a placement position all electrical connection and fixation to the substrate can be made by a conductive material or paste. An example of a conductive adhesives is a thermal or UV settable resin loaded with silver particles, or a solder paste comprising solder particles and flux. Alternatively it may be desired to locally deposit an insulating element, for example to electrically insulate an electric conductor from another, crossing electric conductor. Still further the functional material to be deposited may be a thermal conductor, or instead a thermal insulator.

(6) Printing is an example of a common method for selectively depositing a functional material on a substrate. Typically a functional material needs to be formulated with other materials before the functional material can be printed on a substrate. A formulation can e.g. be formed by dispersing the functional material in a solvent or liquid; thus, the formulation is generally wet. The formulation is often referred to as an ink or paste, depending on the viscosity. A challenge exists to

transfer such functional materials, since ideally, instead of printing a functional ink material on a substrate, selectively depositing only the pure functional material on a substrate is most preferable. Such materials e.g. consist of more than 50 vol % e.g. even 90 vol % of a solid state material such as a conductive polymer. Preferably the functional materials to be deposited are of a high viscosity, e.g. higher than 50 Pa.Math.s to even as much as about 1000 Pa.Math.s (10.sup.6 cp.\$). Provided that shear forces can be avoided during the transfer process, such highly viscous materials will not significantly deform due to gravity and flow, but substantial maintain the intended aspect ratio and form definition in the transfer.

(7) In US20170268100 a transfer method is described, known as photonic printing, using a transparent transfer plate having cavities formed therein, that are filled with a functional material. The transfer plate is irradiated with pulsed light to heat a layer of light-absorbing material in order to generate gas at an interface between the layer of light absorbing material and the functional material, to release the functional material from the cavities. The problem with this method is a non-uniform release of ink due to shadow effects. Especially for high aspect ratio cavities that are closely packed, the light only reaches the sides with difficulty. A non-uniform release of the ink results in poor print qualities, in particular when high resolution patterns are transferred over relatively large distances. Furthermore, to properly transfer feature sizes of 5-10 μm without overheating the ink, the heat flux needs to be at least 40 kW/cm.sup.2 in the order of microseconds, which are unrealistic operation conditions for flashlamps. An other problem is that the method is only suitable for global transfer of structures designed via the patterned cavities provided in the substrate, as the light is not selective. Ideally patterns can be selectively formed and transferred, depending on a required pattern.

(8) Other means are available to provide a heatflux of such magnitudes in a short during pulse. For example, this can be achieved with a laser beam. Generally, a cavity may have dimensions which exceed the dimensions of the spot covered by the laser beam, which necessitates a scanning movement of the laser beam to achieve a transfer of the material over the full length of the cavity.

(9) European patent application EP20171361.7 (P124616EP00), which is not published yet, and filed by the same applicant specifies a method using resistive heating. Therein a resistive heating device in thermal contact with the cavity is provided. The resistive heating device is to heat a functional material in the cavity, having a material composition that, when heated by the heating device, generates a gas at an interface between said cavity surface in the cavity and the functional material. A pressure of the generated gas causes a transfer of the functional material from the cavity onto the receiving substrate.

(10) In case the feature to be deposited on the receiving substrate is relatively large, e.g. in the case of a conducting line to be deposited, it may be complicated to properly control the transfer process. In practice it may not be possible to avoid shear forces, for example due to spatial variations in the heat flux into the functional material. As a result, in the process of transferring, even highly viscous functional materials tend to deform, so that the morphology of the elements formed of the functional material tends to differ from the morphology of the functional material in the cavity in the transfer plate. This deformation could be minimized by reducing the printing gap. However, in many industrial applications, the target substrate on which the functional material is to be deposited often shows substantial height variations, for example due to components mounted on the target. In that case elements deposited on zones of the target surface more remote from the transfer plate surface would be subject to a larger deformation than elements which are closer to the transfer plate surface. To mitigate the effects of these variations on the morphology of the printed elements, industrial application typically require a large printing gap. However, in that case, variations occurring at the time of ejection of the functional material, e.g. due to shear forces are strongly magnified.

(11) Accordingly, there is a need for measures that enable a better of control of the deposition process.

SUMMARY

(12) According to a first aspect of the present disclosure a transfer method is provided as claimed in claim 1. The method as claimed therein comprises the following steps: providing a plate having a donor surface that includes at least one donor area; providing a plurality of individually addressable resistive heater elements in thermal contact with respective zones of the at least one donor area; providing viscous functional material (also denoted as donor material) in the at least one donor area, having a material composition that, when sufficiently heated, generates a gas at an interface between said donor surface in the donor area and said functional material, to transfer said functional material from said at least one donor area by said gas generation onto the receiving substrate; heating respective portions of said viscous functional material in respective zones of the at least one donor area by supplying respective ones of said plurality of individually addressable heater elements with an electric power having a respective time dependent magnitude.

(13) The claimed method renders it possible to better control a heat distribution as well as a distribution of forces exerted on the functional material over the range of the donor area as a function of time. Therewith the process of transferring functional material can be better adapted to the morphology of the target surface.

(14) In some embodiments the donor area is at level with the surrounding surface. This is advantageous in that patterning of the plate is obviated. A printing method, such as screen printing can be used to deposit the functional material according to a desired pattern onto the donor surface. In other embodiments the at least one donor area is formed as a cavity in the donor surface. A higher spatial accuracy can be achieved when the functional material to be deposited is confined in a cavity. In exemplary embodiments a cavity forming a donor area is 1-1000 micron deep and has a smallest diameter ranging between 1 and 5000 micron. The functional material can be provided in the cavity for example using a doctor's blade. In the remainder of the description it will be presumed that the donor area comprising the functional material is formed as a cavity in the donor surface. The present disclosure however equally applies to methods and devices wherein the functional material is provided in a donor area at level with the surrounding surface. Furthermore embodiments may be contemplated wherein some of the donor areas are formed as a cavity and other donor areas are at level with the surrounding surface.

(15) In an embodiment said heating is performed in a plurality of heating cycles, wherein during mutually different phases within a heating cycle a respective one of at least a subset of the plurality of individually addressable heater elements is provided with an electric power, wherein the electric energy provided within a single phase is below a threshold value at which a transfer of the functional material occurs, and the total amount of electric energy provided in said plurality of heating cycles exceeds said threshold value. Accordingly, the plurality of heating cycles comprise initial heating cycles wherein the functional material is gradually heated while avoiding that generated gas causes a transfer of functional material and a final heating cycle wherein sufficient gas is generated to actually transfer the material. In this embodiment an adherence of the functional material in the at least one cavity is minimized during the initial heating cycles. Therewith shear forces exerted on the functional material in the final heating cycle are reduced to a minimum. Therewith the functional material in the at least one cavity can be deposited with minimal morphological changes even at a large distance.

(16) In other embodiments the improved method is used to deliberately adapt the morphology of the functional material to that of the target surface. In some of these other embodiments the plurality of individually addressable heater elements are arranged in successive length zones of an elongate cavity and heating is performed by subsequently supplying electric power to a respective one of the plurality of individually addressable heater elements, wherein a non-linear relationship exists between a rank of the individually addressable heater elements in said succession and a point in time at which a supply of electric power is initiated thereto. In one example thereof the improved method is used to transfer functional material from the at least one cavity at an end portion of a

receiving substrate having an upper target surface, a side target surface and a lower target surface. In this exemplary embodiment of the improved method the cavity includes at least a first length zone, a one but last length zone and a last length zone, wherein heating the last length zone is postponed with a delay period after said heating the one—but last length zone. In a first phase the material from the first length zone is transferred and adheres to the upper target surface. In one or more subsequent phases functional material is ejected from one or more subsequent zones, including the one but last length zone. After these one or more subsequent phases the strip of functional material is attached at one end to the upper target surface and at the opposite end it is still contained in the last length zone of the at least one cavity. The delay period serves as a relaxation time wherein vibrations of the strip of functional material are allowed to dampen out. If these vibrations are sufficiently dampened, the last length zone is heated to eject the strip of functional material at that location. As a result, the strip of functional material folds around the side target surface and the portion originating from the last length zone adheres to the lower target surface. In this exemplary embodiment, the postponement of heating avoids that the strip of functional material hits the side target surface with an excessive velocity, and therewith mitigates the risk that the strip is broken into pieces.

(17) In an embodiment of the method the plate and the receiving substrate are moved relative to each other while in the process of transferring functional material from the at least one cavity, a first portion thereof already adheres to the receiving substrate while another portion thereof still adheres to the plate surface in the cavity. This embodiment provides for various additional processing options. For example, a strip of functional material that is transferred from the cavity to the receiving substrate can be stretched so that a resulting deposited conductor is longer and thinner than the original strip. This implies that for long thin lines, the release of the functional material should be slower than the relative movement of the substrate. As an other example, this embodiment renders it possible to deposit an electrically conductive line that bridges a recession in the substrate surface. I.e. therewith it is avoided that the material of functional material enters the recession.

(18) According to a second aspect of the present disclosure, transfer device is provided for transferring of a viscous functional material, onto a receiving substrate. The transfer device comprises: a plate having a donor surface that includes at least one donor area with said functional material, said functional material having a material composition that, when sufficiently heated, generates a gas at an interface between said donor area surface in the donor area and said functional material, to transfer said functional material from said at least one donor area by said gas generation onto the receiving substrate; a plurality of individually addressable resistive heater elements in thermal contact with a respective zone of the at least one donor area, control circuitry configured to supply an electric power having a respective time dependent magnitude to respective ones of said plurality of individually addressable resistive heater elements.

(19) As noted above, a donor area may be at level with the surrounding surface or may be formed as a cavity in the donor surface. Also in some embodiments a plate is provided with one or more donor areas at level with the surrounding surface and other donor areas formed as a cavity.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) These and other aspects are described in more detail with reference to the drawing. Therein:
- (2) FIG. 1 schematically shows an embodiment of the improved transfer device;
- (3) FIG. 1A shows in more detail elements of an improved transfer device;
- (4) FIG. 2 schematically shows a first embodiment of the improved transfer method;
- (5) FIG. 3 schematically shows a second embodiment of the improved transfer method;

- (6) FIG. 4A-4H shows individual steps of the second embodiment of FIG. 3 in more detail;
- (7) FIG. 5 shows a third embodiment of the improved transfer method;
- (8) FIG. 6A-6F shows individual steps of the third embodiment of FIG. 5 in more detail;
- (9) FIG. 7 shows a fourth embodiment of the improved transfer method;
- (10) FIG. 8A-8E shows individual steps of the fifth embodiment of the improved transfer method;
- (11) FIG. 9A-9C presents examples of resistive heater elements in an embodiment of the transfer device;
- (12) FIG. 10 shows a sixth embodiment of the improved transfer method;
- (13) FIG. 11A, 11B, 11C shows examples of further embodiments of the improved transfer method.

DETAILED DESCRIPTION OF EMBODIMENTS

- (14) Like reference symbols in the various drawings indicate like elements unless otherwise indicated.
- (15) FIG. 1 schematically shows a transfer device **1** for transferring of a viscous functional material **50**, e.g. an electrically conducting material, onto a receiving substrate **20**, such as a circuit board having electronic components to be electrically interconnected. The transfer device **1** comprises a plate **10** that has a cavity surface **11** with at least one cavity **12** filled with the functional material **50** to be transferred. The functional material **50** has a material composition that, when sufficiently heated, generates a gas at an interface between the cavity surface **11** and the functional material **50**. The resulting gas pressure therewith causes a transfer of the functional material **50** from the cavity **12** onto the receiving substrate **20**. In the embodiment shown, the plate **10** comprises a carrier layer **15** and a cover layer **16** of which the latter has a low thermal conductivity to restrict a heat flow to the surface **11** outside the range of the (at least one) cavity **12** and an encapsulation layer **17** of an electrically insulating material. The transfer device **1** comprises a plurality of individually addressable resistive heater elements **131, 132, 133, 134, 135** that are in thermal contact with a respective zone **121, 122, 123, 124, 125** of the cavity **12**. The resistive heating elements are accommodated electrically insulated from each other in the encapsulation layer **17** between the carrier layer **15** and the cover layer **16**. For illustration purposes the size of the heating elements as indicated in the drawings is exaggerated, so that they seem to protrude the carrier layer **15**. FIG. 1 further shows that the surface **11** inside the cavity **12** is coated with a high thermally conductive heat spreader **18** to evenly distribute the heat flux around the functional material, for a more uniform ink release and reducing shear forces. The range within which the heat is spread can be controlled by a proper selection of the thickness of the heat spreader, so that on the one hand a good uniformity is achieved while still being capable to control the distribution of the heat flux at the granularity of the resistive heating elements.
- (16) In the example shown in FIG. 1 the cavity **12** has an elongate form, and the zones **121, 122, 123, 124, 125** are subsequent length zones therein.
- (17) The individually addressable resistive heater elements **131, 132, 133, 134, 135** are coupled to a control circuitry **14**. The control circuitry **14** is configured to supply an electric power **C131, C132, C133, C134, C135** having a respective time dependent magnitude to respective ones **131, 132, 133, 134, 135** of the plurality of individually addressable resistive heater elements.
- (18) Some elements are shown in more detail in FIG. 1A, which shows a portion of a plate **10** comprising an individually addressable heating element **13** which is in thermal contact with a first and a second cavity **12A, 12B**, each provided to transfer a respective portion **50A, 50B** of functional material.
- (19) In some embodiments the device may have a predetermined configuration to operate in a predetermined manner.
- (20) In the embodiment shown, the control circuitry **14** is configured to read a record carrier **60** that comprises instructions that specify the time dependent magnitude according to which it is to supply power to the respective resistive heater elements **131-135**. Mutually different record carriers may specify mutually different embodiments of methods to be performed with the device **1**. Examples

thereof are provided below. Also it may be conceived that the device has a proper record carrier, that can be loaded with instructions for each case, for example instructions received from the internet **61**, or provided by an operator via a user interface **62**.

(21) The substrate **20** may be of a rigid or of a flexible nature. The term “flexible substrate” used in this text refers in particular to a substrate that is bendable enough to be used in a reel to reel process. In other words a flexible substrate as used in this text is a substrate that is flexible enough to allow bending over a certain curvature, e.g. with a radius of 1-100 centimeters (depending on the reel diameter), without the substrate losing essential functionality. Supply of chip bonding material matter **50** can e.g. be placed on a pad structure or on chip pads.

(22) The functional material to be transferred is for example, a thermosettable isotropic conductive adhesive material with a volume resistivity of typically 400-1000 microOhm.Math.cm. To illustrate the general applicability of the method, an experimental conductive adhesive is transferred consisting of a highly viscous conductive adhesive with a viscosity of 160-180 Pa.Math.s. The conductive adhesive can be provided as homogenous layer of 20-30 micron, in particular, 25 micron thick. The thickness is controlled to be around 25 μm or 50 μm but could be theoretically be of any thickness. For example, a cavity **12** may be in the order of 5-10 micron deep and may have a smallest diameter of less than 50 micron.

(23) FIG. 2 schematically shows a method of operating the transfer device **1**. The lower part of FIG. 2 shows a plate **10** that is substantially similar to the one described with reference to FIG. 1 apart from the fact that in this case the cavity **12** comprises three length zones **121**, **122**, **123**. As in the example of FIG. 5, the length zones **121**, **122**, **123** are in thermal contact with a proper individually addressable resistive heater element **131**, **132**, **133**. A viscous functional material **50** to be deposited on the substrate **20** was provided in the cavity **12**, for example using a doctors blade.

(24) The upper part of FIG. 2 shows how the individually addressable heater elements **131**, **132**, **133** each are provided with an electric power **C131**, **C132**, **C133** having a respective time dependent magnitude. In the example shown, heating is performed in a plurality of heating cycles, in this case three heating cycles **C1**, **C2**, **C3**. During mutually different phases within each heating cycle each time one of the individually addressable heater elements **131**, **132**, **133** is provided with an electric power. As indicated by the thick dashed curves, the electric energy (fluence) provided within a single phase is below a threshold value (Release) at which a transfer of the functional material would occur. However, the total amount of electric energy provided in the three heating cycles exceeds the required threshold value. In the example shown, in each cycle electric energy is provided to the heater element **131** in the first phase, to the heater element **132** in the second phase and to the heater element **133** in the third phase. In the third heating cycle **C3**, the accumulated electric energy exceeds the release threshold in the first length zone **121** during the first phase, in the second length zone **122** during the second phase and in the third length zone during the third phase. Because the accumulated energy at the start of the third cycle **C3** is already close to the release threshold, each zone of the cavity **12** only needs to be heated for a short time. Therewith the mutual differences in the release time of the portions of the functional material is small as compared to the case wherein the portions are heated in a single heating cycle. By reducing the differences in release time deformations of the functional material during the process of transfer are minimized.

(25) As shown in FIG. 3 and FIG. 4A-4H, the improved deposition apparatus further renders it possible to deliberately deform the functional material **50** during its deposition in a controlled manner in order adapt the shape of the functional material to the shape of the surface of the substrate **20** where it is to be deposited. In this example the substrate surface comprises an upper surface **21**, a side surface **22** and a lower surface **23**, and the functional material **50** to be deposited is to be wrapped around an end of the substrate **20**, so that it extends over an end portion of the upper surface **21** over the side surface **22** and over an end portion of the lower surface **23**, for example to electrically connect an elements on the upper surface **21** with an element on the lower

surface **23**.

(26) As shown in FIG. 3 and FIG. 4A-4G this is achieved in that the plurality of individually addressable heater elements **131**, **132**, **133**, **134**, **135** are arranged in successive length zones of the elongate cavity **12** filled with the viscous functional material **50** to be deposited at time **T0** (See FIG. 4A). Heating is performed by subsequently supplying electric power to a respective one of the heater elements, wherein a non-linear relationship exists between a rank of the individually addressable heater elements in succession and a point in time at which a supply of electric power is initiated thereto. In the upper part of FIG. 3 this is illustrated in that during a first time interval **T1**, power **C131** is supplied to a first heater element **131** to heat a first length zone of the cavity **12** (See FIG. 4B). As shown in FIGS. 4C, 4D and 4E, during subsequent time intervals **T2-T4** that succeed each other without a significant interruption, power **C132-C134** is supplied to the subsequent heater elements **132-134** to heat subsequent length zones of the cavity **12**. Then, as shown in FIG. 4F, further supply of power is postponed during a time interval **T45** before power **C135** is supplied to the fifth heater element **135** during a fifth time interval **T5**, as shown in FIG. 4G. As schematically shown in the lower part of FIG. 3 and in FIG. 4F, subsequent to the heating step in **T4**, the strip of functional material **50** is adhered at one end to the upper surface of the receiving substrate **20** and at the other end it is still adhered to the surface **11** of the plate inside the cavity **12**. The time interval **T45**, serves as a relaxation time wherein vibrations of the strip of functional material **50** are allowed to dampen out. If these vibrations are sufficiently dampened, the last length zone is heated to eject the strip of functional material at that location. As a result, at time **T6**, shown in FIG. 4H, the strip of functional material **50** folds around the side target surface **22** of the receiving substrate **20** and the portion originating from the last length zone adheres to the lower target surface **23**. The postponement of heating during the time interval **T45** avoids that the strip of functional material **50** hits the side target surface **23** with an excessive velocity, and therewith mitigates the risk that the strip is broken into pieces.

(27) FIG. 5 and FIG. 6A-6F show a further embodiment of the method, wherein different heating modes are combined to fold a strip of functional material **50** during its deposition on the receiving substrate **20**, so that it forms an element **51** of the functional material that has a height exceeding that of the depth of the cavity from which it was transferred. The upper part of FIG. 5 shows the power signal **C131**, . . . , **C135** for each of the heater elements **131**, . . . , **135**, i.e. the power as a function of time supplied thereto.

(28) As shown in FIG. 6A, at **T0**, the plate **10** is provided having its cavity filled with the strip of viscous functional material **50**. During the subsequent time intervals **T1**, **T2** (See FIG. 6B, 6C respectively) the heater elements **131**, **135** at the mutually opposite ends of the cavity are powered with **C131**, **C135** respectively. The amount of energy supplied therewith is sufficient to generate a gas pressure at an interface between the cavity surface in the cavity and the functional material, to force the functional material from the corresponding end-portions of the cavity. Because the functional material **50** has a center portion **52**, **53**, **54** that is still adhered to the cavity surface, the end portions **51**, **55** of the viscous functional material **50** respectively fold over the centrally arranged portions **52**, **54** respectively.

(29) As further shown in FIG. 6D, during the next time interval **T3**, heating of the heater elements **132** and **134** is performed in three heating cycles **C1**, **C2**, **C3**. During a first phase and a second phase within each heating cycle heater element **132**, **134** is provided with an electric power **C132**, **C134**. The electric energy provided within a single phase is below the threshold value at which the functional material portions **52**, **54** are driven out of the cavity, so that these functional material portions are heated in a substantially symmetric manner. During the last heating cycle the amount of energy supplied to these portions exceeds the threshold value so that they are forced to fold inwards. Because only a modest amount of energy needs to be supplied in this last heating cycle, the duration of each phase therein can be short, so that the ejection of the portion **54** can rapidly follow the ejection of portion **52**, therewith providing for a substantially symmetric fold. Finally, as

shown in FIG. 6E, during time interval T5, the most central portion 53 is heated with heater element 133, so that the folded strip 56 of functional material is transferred towards the receiving substrate 20, and adheres thereto at T6 as shown in FIG. 6F. It is noted that during a time interval T4, heating is interrupted, to allow a completion of the folding of the material portions 52, 54.

(30) FIG. 7 shows an example how even more complex constructions can be formed on a substrate surface. The upper part of FIG. 7 shows the surface 11 of the plate 10 that is provided with mutually separate functional material sections 50A, 50B, 50C in respective cavities. The plate 10 has subsequent resistive heater elements 131, 132, 133, 134, 135, 136, 137 having a length in the direction X. Each heater element is configured to heat a region in a proper range in the direction Y. FIG. 7 further shows on the lower left a series of cross-sections A according to A-A as indicated in the upper part, and in the lower-middle and the lower right respectively a series of cross-sections B, C according to B-B and C-C as indicated in the cross-section A. These series of cross-sections are also indicated as A and B in the upper part of FIG. 7.

(31) The upper one of each series of cross-sections A,B,C indicates the plate 10 as it is initially provided at T0.

(32) During a time interval T1, the resistive heater element 131 is heated. As a result, the portions of the material section 50A present in this range, denoted as AL1 on the left and AR2 on the right are heated to an extent that they are forced out by the vapor pressure locally created in the cavity. As a result, these portions fold inward (Fold 1) similarly as the portion 51 in FIG. 6B. Similarly, during a time interval T2, the resistive heater element 137 is heated. As a result, also the portions AL7, AR7 of the material section 50A present in this range are heated to an extent that they are forced out by the vapor pressure locally created in the cavity and fold inwards (Fold 2).

(33) Analogously to what was described with reference to FIG. 6D, during the next time interval T3, heating of the heater elements 132 and 136 is performed in three heating cycles. During a first phase and a second phase within each heating cycle heater element 132, 136 is provided with an electric power. Therewith the portions AL2, AL6, AR2, AR6 of the material section 50A present in the areas that in thermal contact with these heater elements are heated. Also the portions B2, C6 of the material sections 50B and 50C in these areas are heated. The electric energy provided within a single phase is below the threshold value at which locally present functional material portions are driven out of the cavity, so that these functional material portions are heated in a substantially symmetric manner. During the last heating cycle the amount of energy supplied to these portions exceeds the threshold value. Therewith the portion AL2 carrying the earlier folded portion AL1 thereon is folded inwards (Fold 3). Likewise portion AL6 carrying AL7 folds inwards (fold 6). And similarly this applies to AR2 carrying AR1 and to AR6 carrying AR7. Also the portions B2 and C6 fold inwards and therewith fuse at the ends with which they contact each other.

(34) During T4, heating of the heater elements 133 and 135 is performed in three heating cycles. This has as a result that in the last cycle of T4, the material of section 50B present in the range of resistive heater element 133, as well as the material of section 50C present in the range of resistive heater element 135 is ejected. Therewith the fused functional material originating from sections 50B and 50C is deposited as a single conducting strip 57 on the substrate. Also the portions, e.g. AL3, AL6 of material section 50 within the range of heater elements 133 and 135 are forced outwards, but because material section 50A is still adhered to the plate surface in the region of heater element 134, this has as the effect that the structure of material section is folded further, so that the original portions AL1 AL7 fuse with each other. Likewise portions AR1, AR7 fuse with each other. Therewith a bridge 58 is formed that has a first leg 58L formed by portions AL1 AL2, AL3, AL5, AL6 and AL7 at one side, a second leg 58R formed by portions AR1, AR2, AR3, AR5, AR6 and AR7 at the opposite side and a bridge body 58B formed by a portion A4 of the material section in the range of heater element 134. Because this heater element 134 was not yet activated, the bridge 58 is still suspended with the bridge body to the surface 11 of the plate 10.

(35) Finally during T5 heater element 134 is heated so that the bridge 58 which was folded in the

previous steps is now transferred towards the substrate **20**, where it lands at its first and second leg **58L**, **58R** and with the bridge body **58B** traverses the conducting strip **57** at distance.

(36) A further exemplary application is shown in FIG. **8A-8E**. In this example the improved method is applied to provide an electric connection to an electric terminal **26** of a component **25** on the receiving substrate **20**. FIG. **8A** shows the initial situation at **T0**. In FIG. **8B** it is shown how during a time-interval **T1** a central portion of the functional material **50** is released from the cavity by activating resistive heater element **133**. As shown in FIG. **8C**, during a subsequent time-interval **T2** portions of the functional material **50** neighboring this central portion are released by pulse-wise activating the resistive heater elements **132**, **134**. Therewith the elements **132**, **134** the electric energy provided within a single pulse is below a threshold value at which a transfer of the functional material occurs, but the total amount of electric energy provided in the sequence of pulse provided during the time-interval **T2** exceeds the threshold value. Likewise, as shown in FIG. **8D**, during the time-interval **T3**, end portions of the functional material **50** are released by pulse-wise activating the resistive heater elements **131**, **135**. FIG. **8E** shows that as a result at **T4**, a strip of deposited functional material **50** forms an electric connection with the electric terminal **26** of the component **25**.

(37) In the previous examples it was presumed that the resistive heater elements were provided each with separate contacts. FIG. **9A** shows another example, wherein the resistive heater elements of the plate **10** are arranged in a heater matrix. For clarity other details of the plate **10** are not shown in this figure. The heater matrix comprises a first set of powerlines, denoted as column lines **CL1**, **CL2**, **CL3** arranged in a first planar direction of the plate **10** and a second set of power lines, denoted as row lines **RL1**, **RL2**, **RL3** that are arranged transverse to the column lines. In this example the lines are directed in mutually orthogonal direction, but in other examples another angle may be used, e.g. 60 degrees. By way of example, only 3 column lines and 3 row lines are shown. However, in practice a larger number of lines may be present, for example a few hundreds or thousands of lines. Each of the heater elements is electrically connected to a specific pair of a rowline and a column line. For example heater element **H22** has one terminal connected to rowline **RL2** and another terminal connected to column line **CL2**. Neighboring heater element **H23** has one terminal connected to rowline **RL3** and another terminal connected to column line **CL2**.

(38) Heater elements in the matrix can be individually addressed by connecting their corresponding pair of power lines to a power source. In the example shown element **H22** is addressed in that rowline **RL2** and column line **CL2** are connected to the power source. Nevertheless, when connecting the powerlines **RL2**, **CL2**, also a current may flow via other heater elements, for example following the path via heater element **H32**, line **CL3**, heater element **H33**, line **RL3** and heater element **H23**. Although the current flowing via the neighboring heater elements is substantially slower than that via the addressed element, it may be desired to suppress these leakage currents.

(39) To that end, in one example the resistive heater elements are of a material with a negative temperature coefficient (NTC). The effect of this measure is shown in FIG. **9B**. Therein the continuous curves and the dashed curves respectively indicate the resistance and the temperature of a heater as a function of time. For the addressed pixel the curves are black, and for the neighboring pixels the color gray is used. Due to the fact that initially the resistance of the path through the addressed heater element is lower than the resistance of the path through a sequence of serially arranged neighboring elements the temperature rises more rapidly in the addressed heater element than in the neighboring heater elements. As a result of the negative temperature coefficient therewith the resistance of the addressed heater element is reduced more than the resistance of the neighboring heater elements, so that the current path through the addressed heater element is favored more. Hence current through neighboring heater elements is suppressed.

(40) In another example, illustrated with reference to FIG. **9C**, varistor heater elements are used. As in FIG. **9B**, the black dashed curve indicates the temperature development as a function of time for

the addressed heater element and the gray dashed curve indicates the corresponding information for neighboring heater elements. Due to the fact that the current path through the neighboring elements comprises serially arranged varistors, the supply voltage is not sufficient to render these elements conductive, so that a leakage current through these neighboring elements is suppressed.

(41) FIG. 10 shows a further embodiment of the improved transfer method. Therein the plate 10 and the receiving substrate 20 are moved relative to each other during the process of transferring functional material 50 from the cavity 12. In the example shown a first portion 50a of the functional material 50 already adheres to the receiving substrate 20 while another portion 50b of the functional material still adheres to the plate surface in the cavity 12. As a result, the material 50 is stretched so that the form of the resulting deposited feature is longer and thinner than the form of the functional material in the cavity. The extent to which the material is stretched is determined by the ratio of the speed v with which the plate 10 moves relative to the substrate 20 and the speed with which the functional material 50 is expelled from the cavity in the direction of movement. The plate 10 may for example have respective individually addressable resistive heaters each extending over a range with length L in the cavity, that are subsequently heated with a mutual time delay τ , corresponding to a speed of expelling the functional material 50 equal to L/τ . In that case the functional material is stretched with a factor $v \cdot \tau / L$.

(42) A record carrier 60, as shown in FIG. 1 may also comprise instructions that specify the controller 14 how to control the speed v with which the plate 10 moves relative to the substrate 20 and the speed with which the functional material 50 is expelled from the cavity in the direction of movement.

(43) As shown further in FIG. 10 this embodiment of the transfer method also enables a way of deposition wherein the deposited functional material bridges recessed features 20r.

(44) Other variations to the disclosed embodiments can be understood and by those skilled in the art in practicing the claimed invention, from a study of the drawings, the disclosure, and the appended claims. In the claims, the word “comprising” does not exclude other elements or steps, and the indefinite article “a” or “an” does not exclude a plurality. A single unit may fulfill the functions of several items recited in the claims. The mere fact that certain measures are recited in mutually different dependent claims does not indicate that a combination of these measures cannot be used to advantage. Any reference signs in the claims should not be construed as limiting the scope. In addition, the functional material may have a wide-range of viscosity, possible to transfer (from 50 Pa.s to 1000 Pa.s (non-exhaustive)). Contrary to stencil printing and screen printing it is a non-contact direct write method wherein the distance to the substrate surface is preferably kept higher than 1 micron. The transfer accuracy of the functional ink is dependent on the printing gap. Typically the printing gap needs to be in the same order as the feature size to obtain a good printing accuracy. When a small printing gaps are required, it becomes technically more difficult to align the print head with the sample holder. Therefore it is preferred to control the gap by using spacer pins and press the print head and sample holder together. It is preferred to have at least three degrees (heave, roll, pitch) of freedom between the print head and sample holder. In an example pastes can be formed as follows:

(45) TABLE-US-00001 Solids content 90 ± 5 wt % Material Solids content Purpose Metal (Ag, Cu, Au): 86.5 ± 5 wt % Conductivity Polymer binder 2.5 ± 2 wt % Adhesion and cohesion Fumed silica 1 ± 1 wt % Anti-slump, rheology control Solvent: 10 ± 5 wt % Printability

(46) with viscosities ranging between 100 en 1000 Pa*s. In contrast to conventional screen print paste and inks shear thinning may be minimized e.g. by adding fumed silica to prevent flow of the past on the acceptor substrate. This can also be prevented by curing/drying the ink e.g. by laser irradiation, which can be conveniently realized by a high solids content and low solvent content.

(47) The power required to induce a gas pressure is dependent on the functional material used with the method. As a rule of thumb it may be required that the control circuitry connected to the resistive heating elements configured to heat the a heating element in less than 100 microseconds at

an average power of more than 10 kW/cm². Alternatively, it may be required that the control circuitry allows the resistive heater elements to heat the functional material at a heating rate of at least 5K/microsecond in order to generate gas at the interface to provide for the gas pressure that forces the functional material out of the cavity. In case a direct ejection is not required the heating power is provided in a sequence of cycles as shown for example in FIG. 2.

(48) In addition to controlling the transferring process by supplying individually addressable heater elements with an electric power having a respective time dependent magnitude it is possible to control a heat flux distribution associated with an individual heat by means of its shape, as is illustrated in the following examples in FIGS. 11A, 11B and 11C.

(49) In each of these figures, the lower part shows a cross-section through the plate **10** while functional material is being deposited. The middle part shows in a view in a direction towards the surface **11** of the plate **10**, a cross-section of the individual addressable resistive heating element **13**. The upper part schematically shows the local heat flux (solid line) as a function of the position along the cavity **12** and the associated release time (dashed line).

(50) In the example of FIG. 11A the cross-section of the resistive element **13** increases in a direction from left to right along the cavity **12**. As a result, the resistive element has a resistance and therewith a heat flux that decreases in that direction, Consequently the release time for the material **50** in the cavity **12** is larger for positions more to the right end of the cavity. In the lower part of FIG. 11A this is illustrated in that the unit of functional material is tilted with its left end towards the substrate (not shown) where it will be deposited.

(51) In the second example, shown in FIG. 11B, the cross-section the cross-section of the resistive element **13** is narrows at the ends of the cavity **12**, and is relatively wide in a center portion. As a result, the resistive element has a relatively high resistance at the ends of the cavity as compared to in the center portion of the cavity. Therewith the resistive element provides for a relatively high heat flux at the ends of the cavity **12** as compared to the heat flux generated in between the ends, Consequently the release time for the material **50** at the ends of the cavity **12** is relatively short as compared for that of the remainder of the material **50**. In the lower part of FIG. 11B this is illustrated in that the unit of functional material **50** is curved with its ends towards the substrate (not shown) where it will be deposited.

(52) FIG. 11C shows a third example, wherein the cross-section of the resistive element **13** widens at the ends of the cavity **12**, and has a relatively constant small width in a center portion. As a result, the restive element has a relatively low resistance at the ends of the cavity as compared to in the center portion of the cavity. Therewith the resistive element provides for a relatively low heat flux at the ends of the cavity **12** as compared to the heat flux generated in between the ends, Consequently the release time for the material **50** at the ends of the cavity **12** is relatively long as compared for that of the remainder of the material **50**. In the lower part of FIG. 11B this is illustrated in that the unit of functional material **50** is curved with its ends towards the plate **10**.

(53) While the present invention has been described with respect to a limited number of embodiments, those skilled in the art will appreciate numerous modifications and variations therefrom within the scope of this present invention as determined by the appended claims. For example, as noted above, a donor area may be at level with the surrounding surface or may be formed as a cavity in the donor surface. Also in some embodiments a plate is provided with one or more donor areas at level with the surrounding surface and other donor areas formed as a cavity. For example, according to a modification within the scope of the claims a plate having a donor area formed as a cavity in the donor surface may be replaced with a plate having a donor area at level with the surrounding donor surface.

Claims

1. A transfer device for transferring of a functional material, onto a receiving substrate, the device comprising: a plate having a donor surface that includes at least one donor area that is formed as a cavity in the donor surface and that is filled with the functional material, the functional material being one selected from the group consisting of: an electric conductor, an electric insulator, a thermal conductor and a thermal insulator, and wherein the functional material has a viscosity of at least 50 Pa.Math.s and a material composition that, when sufficiently heated, generates a gas at an interface between the donor area surface in the donor area and the functional material, to transfer the functional material from the at least one donor area by the gas generation onto the receiving substrate; a plurality of individually addressable resistive heater elements, each of the individually addressable resistive heater elements being in thermal contact with a respective zone of the cavity forming the at least one donor area; and control circuitry configured to supply an electric power having a respective time dependent magnitude to respective ones of the plurality of individually addressable resistive heater elements, wherein the resistive heater elements are arranged in a heater matrix comprising: a first set of powerlines arranged in a first planar direction of the plate, and a second set of powerlines arranged in a second planar direction transverse to the first planar direction; and wherein respective heater elements are electrically connected to respective pairs of a powerline of the first set and a powerline of the second set.
2. The transfer device according to claim 1, wherein the functional material has a viscosity of at most 1000 Pa.Math.s.
3. The transfer device according to claim 1, wherein the resistive heater elements are of a material with a negative temperature coefficient or are varistor heater elements.
4. A transfer device for transferring of a functional material, onto a receiving substrate, the device comprising: a plate having a donor surface that includes at least one donor area that is formed as a cavity in the donor surface and that is filled with the functional material, the functional material being one selected from the group consisting of: an electric conductor, an electric insulator, a thermal conductor and a thermal insulator, and wherein the functional material has a viscosity of at least 50 Pa.Math.s and a material composition that, when sufficiently heated, generates a gas at an interface between the donor area surface in the donor area and the functional material, to transfer the functional material from the at least one donor area by the gas generation onto the receiving substrate; a plurality of individually addressable resistive heater elements, each of the individually addressable resistive heater elements being in thermal contact with a respective zone of the cavity forming the at least one donor area; and control circuitry configured to supply an electric power having a respective time dependent magnitude to respective ones of the plurality of individually addressable resistive heater elements, wherein the control circuitry is configured to supply the electric power in a plurality of heating cycles, wherein the control circuitry, during mutually different phases within a heating cycle, provides a respective one of at least a subset of the plurality of individually addressable heater elements with an electric power, wherein the electric energy provided within a single phase is below a threshold value at which a transfer of the functional material occurs, and the total amount of electric energy provided in the plurality of heating cycles exceeds the threshold value.
5. The transfer device according to claim 4, wherein the functional material has a viscosity of at most 1000 Pa.Math.s.
6. The transfer device according to claim 4, wherein the resistive heater elements are arranged in a heater matrix comprising: a first set of powerlines arranged in a first planar direction of the plate, and a second set of powerlines arranged in a second planar direction transverse to the first planar direction; and wherein respective heater elements are electrically connected to respective pairs of a powerline of the first set and a powerline of the second set.
7. The transfer device according to claim 6, wherein the resistive heater elements are of a material with a negative temperature coefficient or are varistor heater elements.

8. A transfer device for transferring of a functional material, onto a receiving substrate, the device comprising: a plate having a donor surface that includes at least one donor area that is formed as a cavity in the donor surface and that is filled with the functional material, the functional material being one selected from the group consisting of: an electric conductor, an electric insulator, a thermal conductor and a thermal insulator, and wherein the functional material has a viscosity of at least 50 Pa.Math.s and a material composition that, when sufficiently heated, generates a gas at an interface between the donor area surface in the donor area and the functional material, to transfer the functional material from the at least one donor area by the gas generation onto the receiving substrate; a plurality of individually addressable resistive heater elements, each of the individually addressable resistive heater elements being in thermal contact with a respective zone of the cavity forming the at least one donor area; and control circuitry configured to supply an electric power having a respective time dependent magnitude to respective ones of the plurality of individually addressable resistive heater elements, wherein the transfer device is configured to move the plate and the receiving substrate relative to each other while in the process of transferring functional material from the at least one donor area, a first portion thereof already adheres to the receiving substrate while another portion thereof remains adhered to the plate surface in the donor area.

9. The transfer device according to claim 8, wherein the functional material has a viscosity of at most 1000 Pa.Math.s.

10. The transfer device according to claim 8, wherein the resistive heater elements are arranged in a heater matrix comprising: a first set of powerlines arranged in a first planar direction of the plate, and a second set of powerlines arranged in a second planar direction transverse to the first planar direction; and wherein respective heater elements are electrically connected to respective pairs of a powerline of the first set and a powerline of the second set.

11. The transfer device according to claim 10, wherein the resistive heater elements are of a material with a negative temperature coefficient or are varistor heater elements.

12. A transfer device for transferring of a functional material, onto a receiving substrate, the device comprising: a plate having a donor surface that includes at least one donor area that is formed as a cavity in the donor surface and that is filled with the functional material, the functional material being one selected from the group consisting of: an electric conductor, an electric insulator, a thermal conductor and a thermal insulator, and wherein the functional material has a viscosity of at least 50 Pa.Math.s and a material composition that, when sufficiently heated, generates a gas at an interface between the donor area surface in the donor area and the functional material, to transfer the functional material from the at least one donor area by the gas generation onto the receiving substrate; a plurality of individually addressable resistive heater elements, each of the individually addressable resistive heater elements being in thermal contact with a respective zone of the cavity forming the at least one donor area; and control circuitry configured to supply an electric power having a respective time dependent magnitude to respective ones of the plurality of individually addressable resistive heater elements, wherein the plurality of individually addressable heater elements are arranged in successive zones of the at least one donor area, wherein the control circuitry is configured to sequentially supply an electric power to a respective one of the plurality of individually addressable heater elements, and wherein a non-linear relationship exists between a rank of the individually addressable heater elements in the succession and a point in time at which a supply of electric power is initiated to a particular addressed one of the individually addressable heater elements.

13. The transfer device according to claim 12, wherein the successive zones include at least a first zone, a one but last zone and a last zone, wherein the control circuitry is configured to postpone initiating a supply of electric power to the heater element heating the last zone with a delay period after having supplied an electric power to the heater element of the one—but last zone.

14. The transfer device according to claim 12, wherein the zones include at least a first zone, a one but first zone, a central zone, a one but last zone and a last zone, the control circuitry being

configured to sequentially perform: supplying an electric power to the resistive heater elements of the first and the last zone to eject sections of the viscous functional material therein, thus allowing sections corresponding to the first zone and the last zone to fold towards each other; pulse-wise supplying an electric power to the heater elements associated with the one but first zone and the one but last zone to gradually release sections of the viscous functional material in the one but first zone and the one but last zone, thus allowing sections corresponding to the one but first zone and the one but last zone to fold towards each other; and supplying an electric power to the heater element associated with the central zone to eject folded functional material arising from the supplying an electric power to the resistive heater elements of the first and the last zone and pulse-wise supplying an electric power to the heater elements associated with the one but first zone and the one but last zone.

15. The transfer device according to claim 12, wherein the functional material has a viscosity of at most 1000 Pa.Math.s.

16. The transfer device according to claim 12, wherein the resistive heater elements are arranged in a heater matrix comprising: a first set of powerlines arranged in a first planar direction of the plate, and a second set of powerlines arranged in a second planar direction transverse to the first planar direction; and wherein respective heater elements are electrically connected to respective pairs of a powerline of the first set and a powerline of the second set.

17. The transfer device according to claim 16, wherein the resistive heater elements are of a material with a negative temperature coefficient or are varistor heater elements.
